Floss

```c
xor_encrytion.c
C: > Users > CyberYeti > Desktop > xor_encrytion.c
#include <stdio.h>
#include <windows.h>

/* This is a simple XOR decrypt, used primarily to 'hide' strings in the binary */
void decrypt_encrypt(char*src, size_t key, size_t size);

char mutex_name[25] = "\xc3\xff\xf2\xd4\xee\xf5\xf2\xe5\xce\xf2\xe3\xfe";

const unsigned int xor_key = 0x97;

int main(void) {
    HANDLE h_mutex = NULL;

    wchar_t *string1 = L"this is a wide char string";
    char *string2 = "this is an ascii string";
    char *string3 = "\xe3\xff\xfe\xe4\xb7\xfe\xe4\xb7\xf6\xf9\xb7\xf2\xf9\xf4\xe5\xee\xe7\xe3\xf2\xf3\xb7\xe4\xe3\xe5\xfe\xf9\xf

    decrypt_encrypt(mutex_name, xor_key, strlen(mutex_name));
    h_mutex = OpenMutex(SYNCHRONIZE,FALSE,mutex_name);

    if(!h_mutex) {

        h_mutex = CreateMutex(NULL, TRUE, mutex_name);
        decrypt_encrypt(mutex_name, xor_key, strlen(mutex_name));

        printf("%ls\n", string1);
        printf("%s\n", string2);
        printf("%s\n", string3);
        decrypt_encrypt(string3, xor_key, strlen(string3));
```

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/* This is a simple XOR decrypt, used primarily to 'hide' strings in the binary */
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int main(void) {
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    decrypt_encrypt(mutex_name, xor_key, strlen(mutex_name));
    h_mutex = OpenMutex(SYNCHRONIZE,FALSE,mutex_name);

    if(!h_mutex) {

        h_mutex = CreateMutex(NULL, TRUE, mutex_name);
        decrypt_encrypt(mutex_name, xor_key, strlen(mutex_name));

        printf("%ls\n", string1);
        printf("%s\n", string2);
        printf("%s\n", string3);
        decrypt_encrypt(string3, xor_key, strlen(string3));
        printf("%s\n", string3);
```

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text

00000000  50 4B 03 04 14 00 06 00 08 00 00 00 21 00 6A 2E  PK.........!.j.
00000010  C1 4D A3 01 00 00 1B 06 00 00 13 00 08 02 5B 43  ÁM£.........[C
00000020  6F 6E 74 65 6E 74 5F 54 79 70 65 73 5D 2E 78 6D  ontent_Types].xm
00000030  6C 20 A2 04 02 28 A0 00 02 00 00 00 00 00 00 00  l ¢..( .........
00000040  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000050  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000070  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000080  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000090  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000100  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000110  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000120  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000130  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000140  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000150  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000160  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000170  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000180  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000190  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

```
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00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000070  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000080  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000090  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
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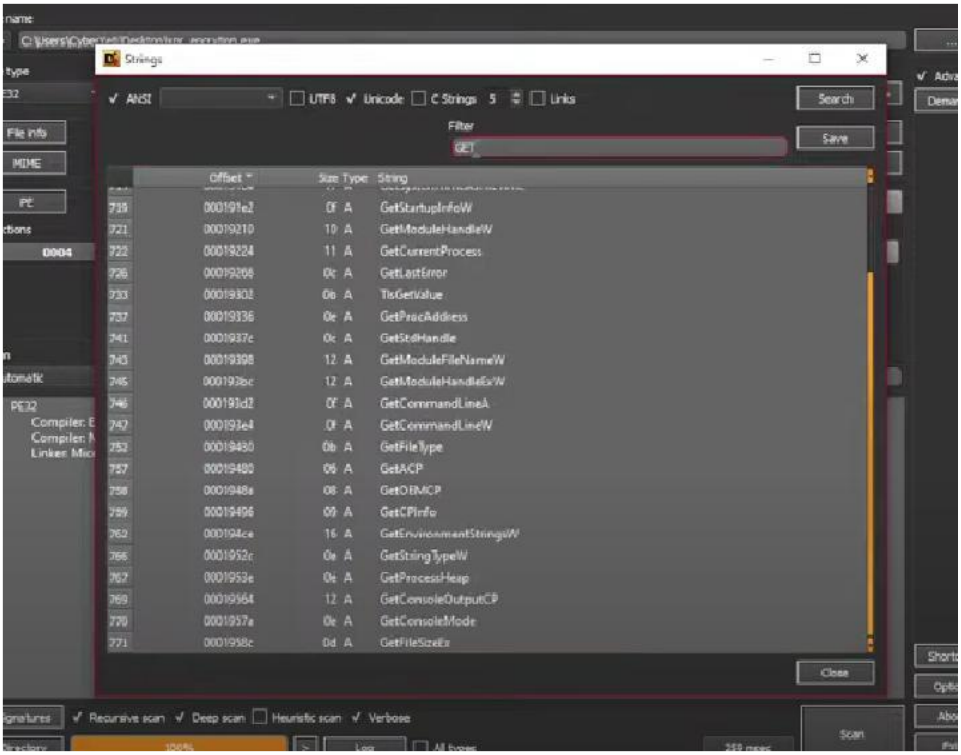
00000160  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000170  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000180  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000190  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000001E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

```
00019550  FlushFileBuffers
00019564  GetConsoleOutputCP
0001957A  GetConsoleMode
0001958C  GetFileSizeEx
0001959C  SetFilePointerEx
000195B0  HeapSize
000195BC  HeapReAlloc
000195CA  CreateFileW
000195D8  WriteConsoleW
000195E8  DecodePointer
00019654  this is an ascii string
0001967A
00019A5A  abcdefghijklmnopqrstuvwxyz
00019A7A  ABCDEFGHIJKLMNOPQRSTUVWXYZ
00019B62
00019C69  abcdefghijklmnopqrstuvwxyz
00019C89  ABCDEFGHIJKLMNOPQRSTUVWXYZ
00019FB8  .?AVbad_exception@std@@
00019FD0  .?AVexcept4on@std@@
00019FF4  .?AVtype_info@@
0001A200  P0W0^0c0q0w0
0001A229  1-1@1J1d1
0001A251  5*51575=5G51505d5y5
0001A24F  6?6Q6
0001A261  7"7(7U7[7
0001A29B  8O9X9]9p9
0001A2C1  ;4;=;L;U;~;
0001A2E1  <&<,<2<8<>=D<K<R<Y<`<g<n<u<|<
0001A319  >7>?>Q>^>
0001A335  1U2a2z2
0001A349  42484n4
0001A35F  5 5.555;5B5c5
0001A377  6?6<6E6d6s6x6|6
0001A39D  7$7)7.7L7Y7h7g717
0001A3B1  7"898
0001A3D5  =4>k>p>t>x>|>
0001A3ED  0&1S1
0001A405  :9:^:=<
0001A41D  0!0;0U0X0d0p0~0
0001A439  1#111<1R1f1o1
0001A447  =!=)=
0001A42B  1&141<1T1m1
0001A4BD  1#2B2
0001A4E1  7%737Y7~7
0001A4F3  8S8:8A8M8a8f8v8|8
```




```
Sysinternals - www.sysinternals.com

No matching files were found.


FLARE-VM Fri 06/21/2024 11:52:03.79
C:\Users\CyberYeti\Desktop>strings.exe xor_encrytion.exe
```

```
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

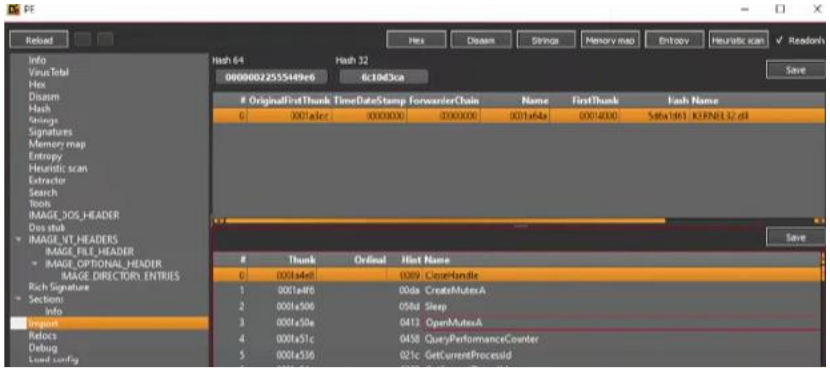
No matching files were found.


FLARE-VM Fri 06/21/2024 11:52:03.79
C:\Users\CyberYeti\Desktop>strings.exe xor_encrytion.exe
```

```
strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

usage: strings.exe [-a] [-f offset] [-b bytes] [-n length] [-o] [-s] [-u] <file or directory>
-a      Ascii-only search (Unicode and Ascii is default)
-b      Bytes of file to scan
-f      File offset at which to start scanning.
-o      Print offset in file string was located
-n      Minimum string length (default is 3)
-s      Recurse subdirectories
-u      Unicode-only search (Unicode and Ascii is default)
-nobanner
        Do not display the startup banner and copyright message.


FLARE-VM Fri 06/21/2024 11:56:32.92
C:\Users\CyberYeti\Desktop>_
```




```
1\1d111t1|1
,242<2D2L2T2\2d212t2|2
,343<3D3L3T3\3d313t3|3
,444<4D4L4T4\4d414t4|4
,545<5D5L5T5\5d515t5|5
,646<6D6L6T6\6d616t6|6
,747<7D7L7T7\7d717t7|7
,848<8D8L8T8\8d8
(30383@3H3P3X3`3h3p3x3
(40484@4H4P4X4`4h4p4x4
(50585@5H5P5X5`5h5p5x5
(60686@6H6P6X6`6h6p6x6
(70787@7H7P7X7`7h7p7x7
(80888@8H8P8X8`8h8p8x8
(90989@9H9P9X9`9h9p9x9
,747<7D7L7T7\7
@>H>L>h>p>t>
$?,?@?H?\?x?
$8(8,8084888<8@8D8P8T8X8\8`8d8h818

RE-VM Fri 06/21/2024 11:57:18.19
Users\CyberYeti\Desktop>
RE-VM Fri 06/21/2024 11:57:46.36
Users\CyberYeti\Desktop>
RE-VM Fri 06/21/2024 11:57:46.45
Users\CyberYeti\Desktop>xor_encrytion
s is a wide char string
```

```
FLARE-VM Fri 06/21/2024 12:02:55.80
C:\Users\CyberYeti\Desktop>floss -h
usage: floss.exe [-h] [-H] [-n MIN_LENGTH] [--no {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]]
                 [--only {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]] [-j] [-v] [-d] [-q] [--color
                 sample

The FLARE team's open-source tool to extract ALL strings from malware.
  floss.exe v3.0.1-0-g3782dc9 - https://github.com/mandiant/flare-floss/

FLOSS extracts the following string types:
 1. static strings:  "regular" ASCII and UTF-16LE strings
 2. stack strings:   strings constructed on the stack at run-time
 3. tight strings:   special form of stack strings, decoded on the stack
 4. decoded strings: strings decoded in a function

Language-specific strings:
 1. Go:   strings from binaries written in Go
 2. Rust: strings from binaries written in Rust

positional arguments:
  sample                path to sample to analyze

optional arguments:
  -h, --help            show this help message and exit
  -H                    show advanced options and exit
  -n MIN_LENGTH, --minimum-length MIN_LENGTH
                        minimum string length

analysis arguments:
  --no {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]
                        do not extract specified string type(s)
  --only {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]
                        only extract specified string type(s)

rendering arguments:
  -j, --json            emit JSON instead of text
  -v, --verbose         enable verbose results, e.g. including function offsets (does not affect JSON output)
```

```
examples:
  extract all strings from an executable
    floss suspicious.exe

  do not extract static strings
    floss --no static -- suspicious.exe

  only extract stack and tight strings
    floss --only stack tight -- suspicious.exe

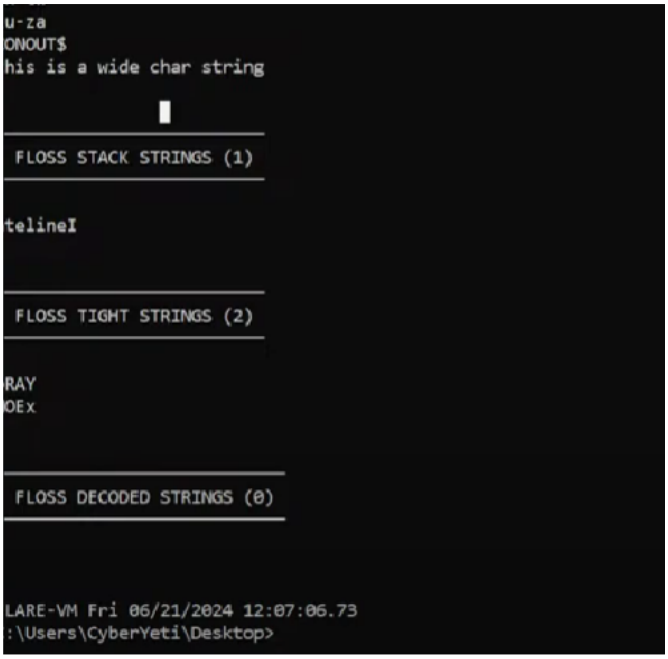
FLARE-VM Fri 06/21/2024 12:03:06.36
C:\Users\CyberYeti\Desktop>floss xor_encrytion.exe
INFO: floss: extracting static strings
finding decoding function features:  29%|[redacted]                    | 179/621 [00:00<00:01,
```

```
examples:
  extract all strings from an executable
    floss suspicious.exe

  do not extract static strings
    floss --no static -- suspicious.exe

  only extract stack and tight strings
    floss --only stack tight -- suspicious.exe

FLARE-VM Fri 06/21/2024 12:03:06.36
C:\Users\CyberYeti\Desktop>floss xor_encrytion.exe
INFO: floss: extracting static strings
finding decoding function features: 100%|[redacted]| 621/621 [00:01<00:00, 358.23 functions/s, skipped 330 library f
INFO: floss.stackstrings: extracting stackstrings from 270 functions
INFO: floss.results: ntelineI
extracting stackstrings: 100%|[redacted]| 270/270 [00:06<00:00, 38.
INFO: floss.tightstrings: extracting tightstrings from 17 functions...
extracting tightstrings from function 0x40cac1:  65%|[redacted]| 11/17 [00:01<00:00,  8.
```



Attribute	Value	Notes
File Name	eicar.com	Standard antivirus test file
MD5 / SHA-256	MD5: 44d88612fea8a8f36de82e1278abb02f	Known EICAR test hash
File Type / Packer	ASCII Text / Not Packed	Simple test file, no packing used
Suspicious Strings	`X5O!P%@AP[4\PZX54(P^)7CC)7}`	Standard EICAR signature string
VirusTotal Detections	60+/70 engines detected	Detected as malicious by multiple AV engines
YARA Rule Name	EICAR_Test_Detection	Rule based on unique string

---

Static Analysis Report

The given sample file was analyzed using **VirusTotal** and basic static analysis techniques. The file was identified as the EICAR test file, which is commonly used to verify antivirus detection systems. The MD5 hash of the file is well-known and flagged by multiple security vendors. The file is not packed and contains a specific signature string that triggers detection. Although it is not harmful, it is treated as malicious by antivirus engines. The analysis identified no complex obfuscation or hidden payloads. A YARA rule was created based on the unique string pattern to detect similar samples.

YARA Rule

```
rule EICAR_Test_Detection
{
    strings:
        $eicar = "X5O!P%@AP[4\\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-
TEST-FILE!"
    condition:
        $eicar
}
```